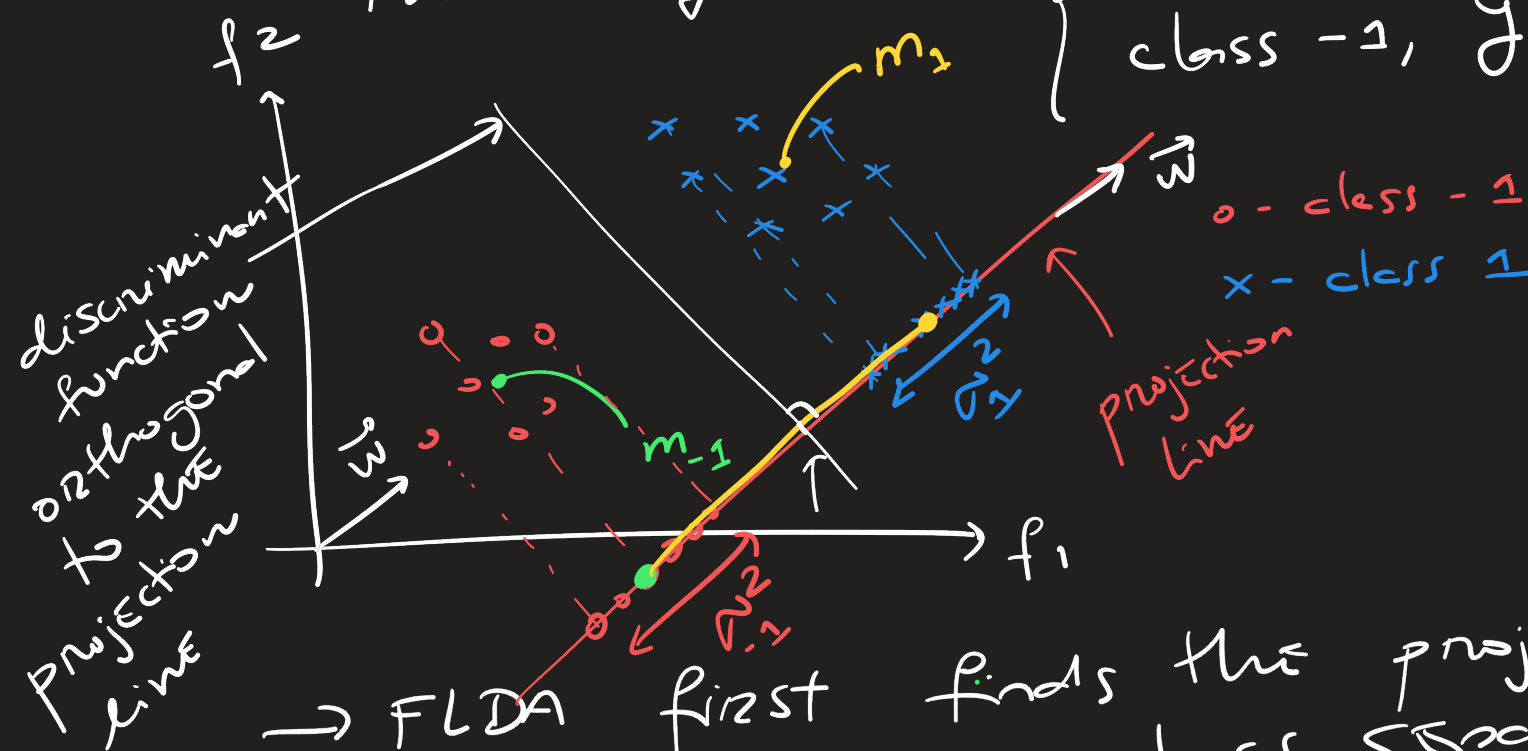


Fisher's Linear Discriminant Analysis (FLDA) OR (LDA).

→ Linear discriminative classifier.

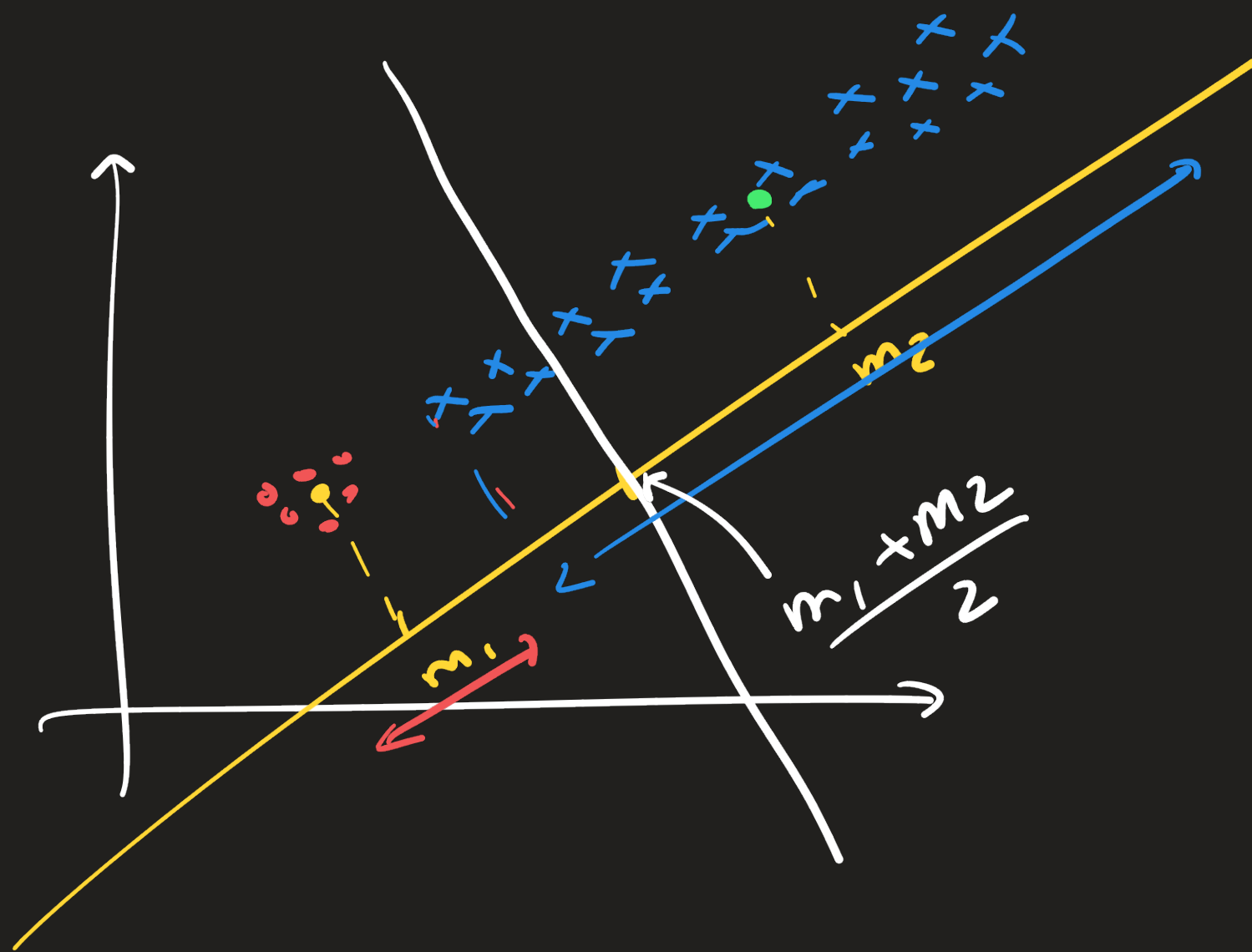
Mapper: $y(x) = w^T x + w_0$.

For binary-class: $\begin{cases} \text{class } 1, y \geq 0 \\ \text{class } -1, y < 0 \end{cases}$



→ FLDA first finds the projection line that

- 1) MAXIMIZES class separability, maximizing $(m_{-1} - m_1)^2$
- and 2) MINIMIZE class VARIANCE.



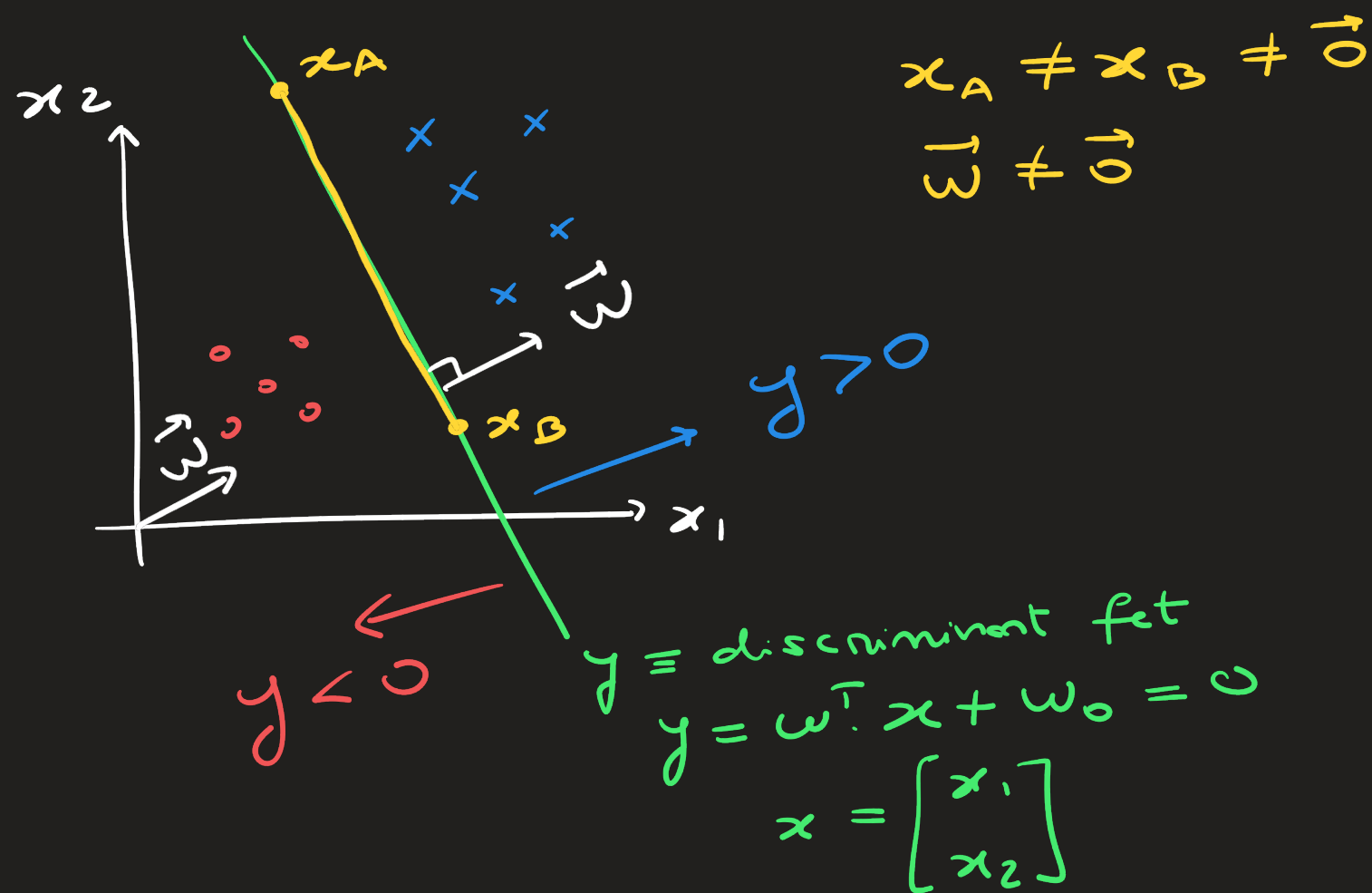
FLDA assumes Gaussian-distributed classes w/ the same isotropic covariances.

$$J(w) = \frac{(m_1 - m_2)^2}{\sigma_1^2 + \sigma_2^2}$$

$m_i \equiv$ average

$\sigma_i^2 \equiv$ variance

for C_i in the projection line
for C_i in the projection line



$$y(x_A) = 0 = y(x_B)$$

$$\vec{w}^T x_A + \cancel{w_0} = \vec{w}^T x_B + \cancel{w_0}$$

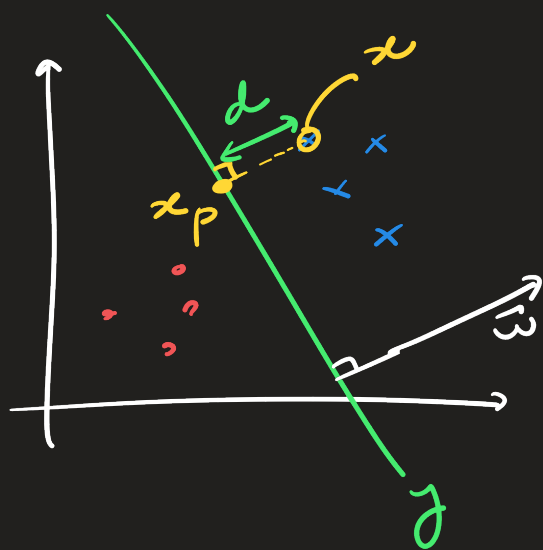
$$\Leftrightarrow \vec{w}^T (x_A - x_B) = 0 \Rightarrow \vec{w} \perp (x_A - x_B)$$

orthogonal or
 perpendicular

Defining "distance" of each point to discriminant f as a function of w .

displacement $\in \mathbb{R}$

$x_p \equiv$ projection of x onto f
 $x \equiv$ original point



$$x = x_p + d \cdot \frac{w}{\|w\|}$$

↑
 was added
 to normalize w
 such that we only
 consider its direction
 not magnitude.

Left-multiply this expression by w^T :

$$w^T \cdot x = w^T x_p + d \cdot \boxed{\frac{w^T w}{\|w\|}}$$

Add w_0 on both sides:

$$\underbrace{w^T x + w_0}_{f(x)} = \underbrace{w^T x_p + w_0}_{f(x_p) = 0} + d \cdot \underbrace{\|w\|}_{\|w\|}$$

$$d = \frac{f(x)}{\|w\|}$$

Objective function for FLDA

$$J(w) = \frac{(m_1 - m_2)^2}{\sigma_1^2 + \sigma_2^2}$$

$m_i \equiv$ projected average for class C_i

$\sigma_i^2 \equiv$ " variance for class C_i

$\vec{m}_i \equiv$ average for class C_i in the original feature space.

$\Sigma_i \equiv$ covariance for class C_i in the original feature space.

$$(m_1 - m_2) = \frac{\omega^T (\vec{m}_1 - \vec{m}_2)}{\|\omega\|}$$

If we assume $\|\omega\| = 1$ (as this can be made possible at the end of our algorithm solution).

$$\underline{(m_1 - m_2)^2} = \omega^T \underbrace{(\vec{m}_1 - \vec{m}_2)(\vec{m}_1 - \vec{m}_2)^T}_{S_B} \cdot \vec{\omega}$$

$S_B \equiv$ between-class
scatter matrix.
($D \times D$)

$$= \omega^T S_B \omega$$

Biased estimator

$$\boxed{\sigma_i^2} = \frac{1}{N_i} \sum_{n \in C_i} (x_n - m_i)^2$$

$$= \frac{1}{N_i} \sum_{n \in C_i} \frac{\vec{w}^T (\vec{x}_n - \vec{m}_i)}{\|\vec{w}\|}$$

Assume $\vec{w}$
 $\|\vec{w}\| = 1$

$$= \frac{1}{N_i} \sum_{n \in C_i} (\vec{w}^T \vec{x}_n - \vec{w}^T \vec{m}_i)^2$$

$$= \frac{1}{N_i} \sum_{n \in C_i} \vec{w}^T (\vec{x}_n - \vec{m}_i) (\vec{x}_n - \vec{m}_i)^T \vec{w}$$

$$= \vec{w}^T \left[\frac{1}{N_i} \sum_{n \in C_i} (\vec{x}_n - \vec{m}_i) (\vec{x}_n - \vec{m}_i)^T \right] \vec{w}$$

$\Sigma_i \equiv$ covariance
 class C_i $\mathbb{R} \times \mathbb{R}$

$$\boxed{= \vec{w}^T \Sigma_i \vec{w}}$$

$$a_1^2 + a_2^2 = w^T \Sigma_1 w + w^T \Sigma_2 w$$

$$= w^T \underbrace{(\Sigma_1 + \Sigma_2)}_{S_w} w$$

$S_w \equiv$ within-class
scatter matrix

$$= w^T S_w w$$

$$J(w) = \frac{w^T S_B w}{w^T S_W w}$$

[quotient rule]

$$\frac{\partial J}{\partial w} = 0 \Leftrightarrow \frac{(w^T S_W w) \cdot \cancel{2} S_B w - (w^T S_B w) \cdot \cancel{2} S_W w}{(w^T S_W w)^2} = 0$$

$$\Leftrightarrow \frac{S_B w}{w^T S_W w} - \boxed{\frac{w^T S_B w}{(w^T S_W w)}} \cdot \frac{S_W w}{(w^T S_W w)} = 0$$

// (unknown)
 $\lambda \equiv \text{constant}$

$$\Rightarrow (S_B) w = (\lambda \cdot S_W) w$$

$$\Rightarrow \boxed{(S_W^{-1} S_B)} w = \boxed{\lambda} \underline{w}$$

w is eigenvector of $(S_W^{-1} S_B)$
w/ Eigenvalue λ .

largest